

Shayan Jalalipour

Portland OR | 503-442-3619 | shayan.jalalipour@gmail.com | [Linkedin](#) | [Github](#) | [Personal Site](#)

Summary

Skilled professional with advanced skills in machine learning and computer science research, working primarily as a research assistant. Developed and applied cutting-edge algorithms to enhance projects in computer vision and generative models, achieving recognition with published state-of-the-art research. Aims to leverage expertise in machine learning and generative modelling techniques to drive innovative solutions and operational success in highly technical environments.

Education

Portland State University 2022 - 2026

PhD, Computer Science with a focus in Machine Learning, Reinforcement Learning, and Computer Vision

Portland State University 2020 - 2022

Master's Degree, Computer science major, focus in Machine Learning

Portland State University 2016 - 2020

Bachelor's Degree, Computer science Major

Work experience

Portland State University Jun 2022 - Present

Machine Learning Research Assistant

Conducted NSF-funded interdisciplinary research by applying and evaluating new machine learning algorithms using Python, PyTorch and other advanced ML tools to publish state-of-the-art research in computer vision, generative models, and ML.

Handshake Oct 2025 - Present

Handshake MOVE Fellow

- Reviewed and annotated large language model (LLM) code outputs in machine learning domains to identify, document, and mitigate behaviors compromising scientific validity, ensuring accuracy and reproducibility of model generated code.
- Applied advanced ML and code-review expertise to audit and annotate LLM generated source code. Identified edge case behaviors and reproducibility issues, strengthening safeguards for the scientific reliability of AI-assisted coding systems.

Portland State University Sep 2021 - Present

Teaching Assistant

Facilitated learning in computer science courses such as Reinforcement Learning, Virtual Reality, and Natural Language Processing by providing clear documentation and effective communication, supporting both students and professors.

Vacasa Jun 2019 - Sep 2019

Data Scientist

Collaborated with a team developing data analysis tools, creating and working with data pipelines, automating geospatial data analysis, as well as researching further applications of machine learning and AI algorithms.

Portland State University Sep 2016 - Sep 2017

IT User Support

Part of Maseeh college of engineering IT empowering user productivity in linux / ubuntu / redhat / windows environments. Supporting students, faculty, and staff with network connectivity and management, account security, and miscellaneous IT needs.

Skills

- Languages:** Python, SQL, C++, C, Java, Javascript, Scala, Prompt Engineering
- Libraries and Frameworks:** Pytorch, CUDA, Tensorflow, Pandas, NumPy, Kubernetes, Docker, SKLearn, HuggingFace, GNNs
- Platforms and Services:** Git, GCP, AWS, GIS, MySQL, MongoDB, OS-terminal coding
- Techniques & Expertise:** Machine Learning, NLP, Statistical Knowledge, Data Visualization, Computer Vision, Deep Learning, Node.js, Transformers, Diffusion Models, Clustering, Multi-Modal Models, Reinforcement Learning, Large Language Models, LLMs, Data Mining

Publications

- Deep Learning-Based Spatial Detection of Drainage Structures Using Advanced Object Detection Methods..2023 *Fifth International Conference on Transdisciplinary AI (TransAI)* <https://ieeexplore.ieee.org/document/10387628>
- Noisy-Defense Variational Auto-Encoder (ND-VAE): An Adversarial Defense Framework to Eliminate Adversarial Attacks..2023 *Fifth International Conference on Transdisciplinary AI (TransAI)* <https://ieeexplore.ieee.org/document/10387596>
- OSA-Diff: An Origin Sampling Based Adversarial Attack Using Diffusion Models..2025 *19th International Conference on Semantic Computing (ICSC)* <https://ieeexplore-ieee-org.proxy.lib.pdx.edu/document/11036312>
- Understanding Adversarial Vulnerabilities and Emergent Patterns in Multimodal RL..2026 *International Conference on Semantic Computing (ICSC)* <https://ieeexplore.ieee.org/document/11486392>